Physical Therapy in Sport 61 (2023) 73–81

Contents lists available at ScienceDirect

Physical Therapy in Sport

journal homepage: www.elsevier.com/ptsp

How are hamstring strain injuries managed in elite men's football clubs? A survey with 62 Brazilian physical therapists

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ARTICLE INFO

Article history:
Received 19 January 2023
Received in revised form
2 March 2023
Accepted 3 March 2023

Handling Editor: Dr L Herrington

Keywords:
Soccer
Muscle injury
Physical therapy
Rehabilitation

ABSTRACT

Objective: To describe perceptions and practices of physical therapists from elite men's football clubs on the management of athletes with hamstring strain injury (HSI).
Design: Cross-sectional study.
Setting: Online survey.
Participants: Physical therapists from clubs engaged in the two main divisions of Brazilian men's football.
Main outcome measures: Practices for assessment and rehabilitation of athletes with HSI.
Results: This survey had 62 physical therapists from 35 of the 40 eligible clubs (87.5% representativeness). Despite heterogeneity on assessment practices, all respondents use imaging exams, adopt injury classification scales, and evaluate aspects related to pain, range of motion, muscle strength, and functional status of athletes with HSI. Rehabilitation programs are usually divided into 3 to 4 phases. All respondents usually apply electrophysical agents and stretching in HSI rehabilitation programs, 98.4% apply strengthening exercises (93.5% include eccentrics), 96.8% manual therapy, 95.2% exercises that mimic the functional demands of football, and 93.5% lumbopelvic stabilization exercises. Muscle strength was the most reported return to play criterion (71% of respondents).
Conclusion: The present study allowed the sports physical therapy community to become aware of the approaches usually adopted for management of athletes with HSI who play in the highest level of Brazilian men's football.

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1. Introduction

The hamstring strain injury (HSI) is the most prevalent non-contact injury in professional men's football (soccer), representing near to 19% of all injuries (Ekstrand et al., 2020, 2022). Throughout a season, elite clubs can expect that one out of five athletes in a team squad miss training or game due to HSI (Ekstrand et al., 2022). This injury recurrence rate is high (around 18%) (Ekstrand et al., 2020, 2022), and re-injuries usually cause a longer time-loss than index injuries (mean of 21.5 and 18 days, respectively, for structural injuries) (Ekstrand et al., 2020). The HSI rate has increased since the beginning of this century (Ekstrand et al., 2022), and this injury burden has a negative impact on club's performance and finances (Eliakim et al., 2020; Hägglund et al., 2013). Therefore, the health staff work hard to return injured athletes to the field as early as possible, with the greatest possible conditioning level and with the lowest possible recurrence risk.

A series of randomized clinical trials (RCTs) (Askling et al., 2013, 2014; Bayer et al., 2017; Hickey et al., 2020; Malliaropoulos et al., 2004; Medeiros et al., 2020; Mendiguchia et al., 2017; Sherry & Best, 2004; Silder et al., 2013; Vermeulen et al., 2022) and cohort studies (Hickey et al., 2022; Jacobsen et al., 2016; Moen et al., 2014; Shamji et al., 2021; Tol et al., 2014; Wangensteen et al., 2015; Whiteley et al., 2018) conducted by different groups worldwide have provided commendable advances on management of athletes with HSI in the last two decades, culminating in the recent publication of a 'Clinical Practice Guideline' (Martin et al., 2022) on this topic. A range of methods have been used to assess pain, range of motion, muscle strength, and functional deficits in athletes with HSI; and there is no consensus on classification/grading of muscle injuries (Hamilton et al., 2015) or prognostic value of imaging

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https://doi.org/10.1016/j.ptsp.2023.03.001
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exams (Jacobsen et al., 2016; Shamji et al., 2021; Wangensteen et al., 2015). During the rehabilitation program, physical therapists have applied passive strategies such as manual therapy (Jacobsen et al., 2016; Mendiguchia et al., 2017; Tol et al., 2014; Vermeulen et al., 2022; Whiteley et al., 2018) and electrophysical agents (Malliaropoulos et al., 2004; Medeiros et al., 2020; Mendiguchia et al., 2017; Sherry & Best, 2004; Silder et al., 2013), but exercise is considered the primary intervention (Martin et al., 2022). Specific strengthening of the hamstrings has been the most adopted approach, specially including eccentric exercises, often added to exercises targeted on lumbopelvic stabilization and agility, as well as progressive running protocols (Afonso et al., 2023; Breed et al., 2022; Martin et al., 2022). Lastly, decision-making on return to play (RTP) is challenging and a range of criteria such as pain, range of motion, muscle strength, functional status, and psychological readiness have been recommended (van der Horst et al., 2017).

It is noteworthy that most studies have not been conducted in the 'real world' of elite men's football, which has a specific biopsychosocial context. The club's facilities, human and material resources available for treatment, physical/technical demands required for RTP, and external pressure (i.e., coach, board, sponsors, press, fans), among other factors, play a role on athletes' rehabilitation. Consequently, a successful rehabilitation program supported by a study with recreational athletes, practitioners of football or other sports, available for only 2–3 weekly rehabilitation sessions, and without the premise of RTP as early and fittest as possible cannot be directly replicated into an elite football club. Despite RTP practices adopted by elite football clubs following HSI have already been described in a worldwide survey (Dunlop et al. 2020), it is unclear how physical therapists actually proceed with the assessment and rehabilitation of such cases. Therefore, the aim of this study was to describe perceptions and practices of physical therapists from elite men's football clubs on the management of athletes with HSI.

2. Methods

2.1. Study design

This is a cross-sectional study. The participants answered a questionnaire about their perceptions and practices in the management of professional football athletes with HSI. The study was approved by the institutional ethics in research committee (#4.838.243) and all participants signed the consent form.

2.2. Participants

According to official data provided by *Fédération Internationale de Football Association* (FIFA), Brazil is the country with the highest number of men's football clubs worldwide, counting with 656 professional clubs (FIFA Landscape, 2023). The Brazilian Championship is the main national competition, and elite clubs play in the two top tiers, named as Series A (first division, 20 clubs) and Series B (second division, 20 clubs). The current study had the participation of physical therapists from the clubs that competed in the Brazilian Men's Football Championship Series A and Series B at the 2021 season. Considering an average of 3–4 physical therapists per club, the population was estimated in approximately 140 individuals. To guarantee the representativeness of this population, it was stipulated that ¾ of the total clubs (i.e., 30 clubs) should have at least one physical therapist composing the sample of the current study.

2.3. Procedures

Since no studies were found with a similar purpose to the present study, a specific questionnaire was created by authors: a physical therapist with a 20-year career within an elite football club (H.G.V.); a PhD university professor with previous experience as head physical therapist of an elite football club (R.R.O); and a PhD university professor familiarized with research in the football setting (B.M.B.). Next, two physical therapists not involved in this survey and with experience working in elite men's football clubs were asked to review the questionnaire to verify cultural issues and adequacy of language to the participants. The suggestions of these external consultants for changing some terms were discussed among the authors and the necessary changes were made when considered appropriate.

The original questionnaire was developed and delivered in Brazilian Portuguese to ensure the full understanding of the participants. In addition to participants' demographic aspects, the final version of the questionnaire included 30 questions, divided into 3 sections: (i) questions about the structure and organization of the physical therapy section of the club where the participants worked; (ii) questions about the physical therapy assessment of athletes with HSI; and (iii) questions about the rehabilitation program of athletes with HSI.

The researchers contacted at least one physical therapist from each of the 40 clubs eligible for the study, who assisted in obtaining the contacts of other physical therapists from their respective clubs. The invitation to participate in the study was sent via WhatsApp Messenger (Meta, California, USA). Physical therapists who demonstrated interest in participating received more details about the study procedures, as well as the guarantee of confidentiality of responses. The participants' doubts were solved before signing the consent forms. Both consent forms and questionnaires were sent to participants via WhatsApp Messenger or email, according to the participant's preference.

Data collection took place between March and December 2021. A flexible deadline was established for participants to send the questionnaire fully answered. Consequently, the return occurred in periods that varied from a few days to a few weeks. As soon as each questionnaire was received, researchers reviewed all responses. When unanswered questions or questions with dubious answers were identified, a researcher (H.G.V.) contacted the participant for clarification.

2.4. Data analysis

Raw data were exported and analyzed on Microsoft Excel software. Exploratory data analysis was performed to determine the distribution of participants' responses, as well as mean and standard deviation for demographic data.

3. Results

Ninety-five physical therapists from the 40 clubs engaged in Brazilian Men's Football Championship 2021, Series A and Series B, were invited to participate in the study. Sixty-two physical therapists agreed to answer the survey and returned the questionnaires within the stipulated period (65% adherence). Participants worked in 35 of the 40 clubs eligible for this study (87.5% representativeness), being 17 clubs engaged in Series A and 18 in Series B. All participants were men and have born in Brazil. Further demographic data are presented in Table 1. Details regarding structure and organization of their workplaces are presented in Table 2.

Perceptions and practices of physical therapists on the assessment and rehabilitation of athletes with HSI are presented in

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Table 1
Characteristics of participants.

Demographic data	mean ± SD (Min; Max) or distribution
Age (years)	38.0 ± 6.8 (26; 55)
Experience as a physical therapist (years)	13.1 ± 6.8 (1; 30)
Experience with professional football (years)	9.5 ± 6.7 (0.5; 26)
Experience in the current professional team (years)	7.0 ± 6.4 (0.25; 26)
Professional/Academic degrees (distribution)	
- SONAFE specialist certification	12 (19.4%)
- Specialization course	46 (74.2%)
- Master's degree	11 (17.7%)
- Doctoral degree	3 (4.8%)

Max, maximum; Min, minimum; SD, standard deviation; SONAFE, Brazilian National Society of Sports Physical Therapy.

Table 2
Characteristics of the participants' workplaces. Results are presented as absolute and percent distribution.

Question	Answer	N (%)
1) In addition to the physical therapy room, what facilities do you have access for rehabilitation activities?	Football field	62 (100%)
	Gym room	62 (100%)
	Pool	37 (59.7%)
	Grass court (natural or synthetic)	28 (45.2%)
	Sand court	26 (41.9%)
	Multi-sport court	7 (11.3%)
2) Regarding electrophysical agents, which ones do you have available in the club?	Cryotherapy	62 (100%)
	Analgesic currents (eg, TENS)	62 (100%)
	Pneumatic compression boots	61 (98.4%)
	Excitomotor currents (eg, NMES)	60 (96.8%)
	Therapeutic ultrasound	57 (91.9%)
	Photobiomodulation (ie, lasertherapy)	56 (90.3%)
	Diathermy (eg, short wave, microwave)	53 (85.5%)
	Compression boots with cryotherapy	53 (85.5%)
	Shock waves	27 (43.5%)
	Magnet therapy	23 (37.1%)
	Intratissue percutaneous electrolysis	22 (35.5%)
	Others	3 (4.8%)
3) Regarding resistance exercise, which resources do you have available in the club?	Free weights (eg, dumbbells, barbells, plates)	62 (100%)
	Weight machines (eg, leg-press, leg curl)	62 (100%)
	Elastic bands	62 (100%)
	Isokinetic dynamometer	42 (67.7%)
	Pneumatic equipment	36 (58.1%)
	Isoinertial equipment	28 (45.2%)
	Others	6 (9.7%)
4) Regarding aerobic exercise, which resources do you have available in the club?	Treadmill	62 (100%)
	Exercise bike	62 (100%)
	Track or outdoor space	60 (96.8%)
	Elliptical	43 (69.4%)
	Others	6 (9.7%)
5) Is there a coordinator in the physical therapy sector?	No	14 (22.6%)
	Yes	48 (77.4%)
6) Is there a physical therapist in the field during training sessions?	Always	25 (40.3%)
	Sometimes	32 (51.6%)
	Never	4 (6.5%)
	**missing data (n = 1)*	
7) Is there a physical therapist in the field during games?	Always	35 (56.5%)
	Sometimes	17 (27.4%)
	Never	10 (16.1%)
8) Do the club's physical therapists follow a standard protocol for assessment of athletes with HSI?	No	11 (17.7%)
	Yes	51 (82.3%)
9) Do the club's physical therapists follow a standard protocol for rehabilitation of athletes with HSI?	No	17 (27.4%)
	Yes	45 (72.6%)
10) Do the club's physical therapists follow a standard protocol for RTP of athletes with HSI?	No	7 (11.3%)
	Yes	55 (88.7%)

HSI, hamstring strain injury; NMES, neuromuscular electrical stimulation; RTP, return to play; TENS, transcutaneous electrical nerve stimulation.

Tables 3 and 4, respectively. There was missing data in some questions (detailed in tables) because a few participants did not respond to the researchers' contact to clarify their dubious answers.

4. Discussion

Given the unsatisfactory results of invasive regenerative therapies such as platelet-rich plasma injection (Seow et al., 2021) and the lack of high-level clinical studies with pharmacological agents (eg, non-steroidal anti-inflammatory drugs and corticosteroids), physical therapists have assumed the leading role on management of athletes with HSI (Martin et al., 2022). Interestingly, the present study is the first to describe assessment procedures and rehabilitation programs following HSI implemented by physical

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Table 3
Perceptions and practices of physical therapists on the assessment of athletes with hamstring strain injury. Results are presented as absolute and percent distribution.

Question	Answer	N (%)
1) How long after the injury event do you believe is the best time to evaluate athletes with HSI?	Immediately after the injury	11 (17.7%)
	First 12 h after injury	18 (29%)
	12–24 h after injury	22 (35.5%)
	24–48 h after injury	11 (17.7%)
	More than 48 h after the injury	0
2) Do you usually have imaging exams of athletes with HSI?	No	0
	Yes	62 (100%)
3) How important is it for you to have imaging exams of athletes with HSI?	Very important	46 (74.2%)
	Important	16 (25.8%)
	Little important	0
	Not important	0
4) Do you use any classification system for HSI? If yes, which one(s)?	No	0
	Yes	62 (100%)
	Munich Consensus	28 (45.2%)
	British Classification	15 (24.2%)
	Subjective classification (mild, moderate, severe)	15 (24.2%)
	Classification in degrees (I, II, III)	12 (19.3%)
5) Do you assess pain of athletes with HSI?	No	0
	Yes	62 (100%)
If yes, what parameters are evaluated?	Intensity of pain	58 (93.5%)
	Site of pain	54 (87.1%)
	Extension of the painful zone	48 (77.4%)
If yes, which method(s) do you use?	Numeric rating scale	37 (59.7%)
	Visual analogic scale	39 (62.9%)
	Pressure algometry	3 (4.8%)
If yes, under what conditions is pain evaluated?	Pain on contraction	62 (100%)
	Pain on palpation	61 (98.4%)
	Pain on stretching	55 (88.7%)
	Pain at rest	37 (59.7%)
	Pain during any specific test	17 (27.4%)
6) Do you assess range of motion of athletes with HSI? If yes, which method do you use?	No	1 (1.6%)
	Yes	60 (96.8%)
	**missing data (n = 1)*	
	Quantitative analysis (eg, goniometry)	36 (58.1%)
	Subjective analysis (eg, visual inspection)	35 (56.5%)
7) Do you assess the hamstring strength of athletes with HSI?	No	0
	Yes	62 (100%)
If yes, which method do you use?	Testing against manual resistance	52 (83.9%)
	Testing with hand-held dynamometer	22 (35.5%)
	Testing with load cells	6 (9.7%)
	Isometric	48 (77.4%)
If yes, in what type(s) of muscle contraction?	Concentric	44 (71%)
	Eccentric	39 (62.9%)
8) Do you assess the functional status of athletes with HSI? If yes, how?	No	0
	Yes	62 (100%)
	Basic movements (eg, squat, step up/down)	51 (82.3%)
	Running	29 (46.8%)
	Askling H-test	26 (41.9%)
	Walking	25 (40.3%)
	Jump	16 (25.8%)
9) Do you assess the psychological status of athletes with HSI? If yes, how?	No	1 (1.6%)
	Yes	61 (98.4%)
	During a casual conversation	51 (82.3%)
	Refers to the club psychologist when necessary	16 (25.8%)
	Uses a questionnaire	3 (4.8%)

HSI, hamstring strain injury.

therapists from elite men's football clubs.

While some physical therapists in the current study advocate evaluation immediately after the injury event, others prefer to evaluate the athlete after 24–48 h. Conversely, all respondents considered the imaging exams as 'important' or 'very important', even its prognostic value to determine the time to RTP and the reinjury risk is unclear (Jacobsen et al., 2016; Shamji et al., 2021; Wangensteen et al., 2015). Involvement of tendon tissue increases the time to RTP following HSI (Shamji et al., 2021), but only ¼ of respondents adopt the 'British Athletic Muscle Injury Classification' (Pollock et al., 2014), which allows the identification of those cases with worse prognosis. Approximately 45% of respondents adopt the 'Munich Consensus Classification' (Mueller-Wohlfahrt et al., 2013), either alone or associated with other classifications. The 'Munich Consensus Classification' does not take into account affections to the tendon tissue but provide more comprehensive diagnoses than the traditional grade 1-to-3 classification systems (Hamilton et al., 2015).

In the assessment of pain intensity, numerical rating scale and visual analogic scale share the respondents' preference, who usually perform pain measurements during contraction, palpation and stretching. Despite the inherent subjectivity of such perception scales, its use is widespread not only in the initial assessment but also for pain monitoring throughout rehabilitation (Hickey et al., 2020, 2022; Whiteley et al., 2018). There is a grade 'A' of recommendation (i.e., strong evidence) for the use of the inclinometer

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Table 4
Perceptions and practices of physical therapists on the rehabilitation of athletes with hamstring strain injury. Results are presented as absolute and percent distribution.

Question	Answer	N (%)
1) Do you usually organize the rehabilitation program for athletes with HSI in phases? If yes, how many?	No	5 (8.1%)
	Yes	56 (90.3%)
	**missing data (n = 1)*	
	2 phases	2 (3.2%)
	3 phases	28 (45.2%)
	4 phases	21 (33.9%)
	5 phases	2 (3.2%)
	8 phases	2 (3.2%)
	9 phases	1 (1.6%)
2) Do you usually apply specific criteria for the progression of the rehabilitation program? If yes, which one(s)?	No	1 (1.6%)
	Yes	59 (95.2%)
	**missing data (n = 1)*	
	Pain	58 (93.5%)
	Strength	55 (88.7%)
	Functional status	53 (85.5%)
	Time	45 (72.6%)
	Range of motion	42 (67.7%)
	Psychological status	24 (38.7%)
3) Do you usually apply electrophysical agents in the rehabilitation of athletes with HSI? If yes, which one(s)?	No	0
	Yes	62 (100%)
	Analgesic currents (eg, TENS)	58 (93.5%)
	Photobiomodulation (ie, lasertherapy)	55 (88.7%)
	Cryotherapy	54 (87.1%)
	Compression boots with cryotherapy	46 (74.2%)
	Diathermy (eg, short wave, microwave)	46 (74.2%)
	Pneumatic compression boots	43 (69.4%)
	Therapeutic ultrasound	39 (62.9%)
	Excitomotor currents (eg, NMES)	21 (33.9%)
	Intratissue percutaneous electrolysis	19 (30.6%)
	Magnet therapy	11 (17.7%)
	Shock waves	11 (17.7%)
4) Do you usually apply manual therapy techniques in the rehabilitation of athletes with HSI? If yes, which one(s)?	No	1 (1.6%)
	Yes	60 (96.8%)
	Myofascial release	59 (95.2%)
	Mobilization	47 (75.8%)
	Manipulation	37 (59.7%)
	Massage	33 (53.2%)
	Lymphatic drainage	19 (30.6%)
	Osteopathy	16 (25.8%)
	Maitland	1 (1.6%)
Holfing	1 (1.6%)	
5) Do you usually apply other passive therapies in the rehabilitation of athletes with HSI? If yes, which one(s)?	No	23 (37.1%)
	Yes	39 (62.9%)
	Elastic taping	30 (48.4%)
	Locally applied substances (eg, ointments)	12 (19.4%)
	Rigid taping	2 (3.2%)
	Bracing	2 (3.2%)
	Compressive stockings	1 (1.6%)
	Acupuncture	1 (1.6%)
	Dry needling	1 (1.6%)
6) Do you usually apply exercises to increase range of motion in the rehabilitation of athletes with HSI? If yes, which one(s)?	No	0
	Yes	62 (100%)
	Dynamic stretching	51 (82.3%)
	Static stretching	27 (43.5%)
	Ballistic stretching	27 (43.5%)
	Pilates	12 (19.4%)
	Proprioceptive Neuromuscular Facilitation	11 (17.7%)
	Isostretching	1 (1.6%)
	Muligan	1 (1.6%)
	Maitland	1 (1.6%)
7) Do you usually apply exercises for muscular strengthening in the rehabilitation of athletes with HSI?	No	1 (1.6%)
	Yes	61 (98.4%)
If yes, do you include exercises with eccentric overload?	No	2 (3.2%)
	Yes	58 (93.5%)
	**missing data (n = 1)*	
If yes, what types of exercise do you most commonly use?	Body-weight exercises	55 (88.7%)
		54 (87.1%)

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Table 4 (*continued*)

Question	Answer	N (%)
	Free weights (eg, barbells, plates, shin guards)	
	Exercises on machines (eg, leg curl)	51 (82.3%)
	Isometric exercises	47 (75.8%)
	Exercises with resistance bands	40 (64.5%)
	Isokinetic exercises	32 (51.6%)
If yes, how many sets per exercise do you most commonly use?	1 set	0
	2 sets	5 (8.1%)
	3 sets	40 (64.5%)
	4 sets	28 (45.2%)
	5 sets	3 (4.8%)
	**missing data (n = 2)*	
If yes, what repetition range per set do you most commonly use?	<5 reps	2 (3.2%)
	5-10 reps	33 (53.2%)
	10-15 reps	38 (61.3%)
	15-20 reps	5 (8.1%)
	>20 reps	0
	** missing data (n = 2)*	
8) Do you usually apply lumbopelvic stabilization exercises in the rehabilitation of athletes with HSI? If yes, which one(s)?	No	4 (6.5%)
	Yes	58 (93.5%)
	Static exercises	29 (46.8%)
	With slow/controlled movements	46 (74.2%)
	With explosive movements	26 (41.9%)
	With unexpected stimuli (perturbations)	20 (32.3%)
9) Do you usually apply exercises that mimic functional demands of football in the rehabilitation of athletes with HSI? If yes, which one(s)?	No	3 (4.8%)
	Yes	59 (95.2%)
	Breaking and acceleration	59 (95.2%)
	High-speed turns and changes of direction	58 (93.5%)
	High-speed running	57 (91.9%)
	Jumps and landings	56 (90.3%)
	Passes and kicks	56 (90.3%)
	Sprints	55 (88.7%)
10) Do you normally carry out periodic assessments during the rehabilitation program for athletes with HSI? If yes, which ones?	No	0
	Yes	62 (100%)
	Pain	52 (83.9%)
	Muscle strength	51 (82.3%)
	Functional status	41 (66.1%)
	Range of motion	37 (59.7%)
	Askling H-test	8 (12.9%)
	SLHB test	8 (12.9%)
	Sports gesture on the field	8 (12.9%)
	GPS data	6 (9.7%)
	Imaging exam	5 (8.1%)
	Psychological status	5 (8.1%)
	Thermography	3 (4.8%)
11) Do you adopt specific criteria to release athletes following HSI to return to training with the team (or with the staff responsible for transition)? If yes, which one(s)?	No	2 (3.2%)
	Yes	60 (96.8%)
	Muscle strength	44 (71%)
	Sprints/high-speed running	36 (58.1%)
	Intense actions (eg, accelerations)	27 (43.5%)
	Pain	27 (43.5%)
	External workload (GPS metrics)	20 (32.3%)
	Range of motion	17 (27.4%)
	Functional tests (eg, hops, H-test, SLHB)	17 (27.4%)
	Imaging exam	14 (22.6%)
	Time since injury event	2 (3.2%)

GPS, Global Positioning System; HSI, hamstring strain injury; NMES, neuromuscular electrical stimulation; SLHB, single-leg hamstring bridge; TENS, transcutaneous electrical nerve stimulation.

during evaluation of athletes with HSI (Martin et al., 2022), but 39% of respondents perform only subjective analysis of range of motion. Similarly, there is a grade 'A' of recommendation for objective quantification of muscle strength after HSI (Martin et al., 2022), but 56.5% of respondents apply only manual testing to identify muscle strength deficits. Given that both passive range of motion testing (Moen et al., 2014) and strength testing (Jacobsen et al., 2016) have been associated with the time to RTP following HSI, such findings were somewhat unexpected, especially considering that elite football clubs have resources enough for achieving inclinometers and handheld dynamometers.

Most respondents (90%) organize their HSI rehabilitation program in phases. The number of program phases varies between 2 and 9, with the majority adopting 3 or 4 phases (44% and 34%, respectively). Three-phase programs seem to be the most frequent in the literature (Bayer et al., 2017; Medeiros et al., 2020;

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Mendiguchia et al., 2017; Silder et al., 2013; Vermeulen et al., 2022; Whiteley et al., 2018). The progression of athletes within rehabilitation programs is usually based on clinical and functional criteria, and the present study shows that pain (93.5%), muscle strength (89%) and functionality (85.5%) are the criteria most adopted. Interestingly, 76% of respondents use the time elapsed since injury as a criterion for progression of the rehabilitation program. This criterion has not been adopted in the main RCTs (Askling et al., 2013, 2014; Bayer et al., 2017; Hickey et al., 2020; Malliaropoulos et al., 2004; Medeiros et al., 2020; Mendiguchia et al., 2017; Sherry & Best, 2004; Silder et al., 2013; Vermeulen et al., 2022) nor in the esteemed 'Aspetar Hamstring Protocol' (Jacobsen et al., 2016; Tol et al., 2014; Wangensteen et al., 2015; Whiteley et al., 2018), but there is a contemporary debate about respecting time for natural healing of HSI (Pieters et al., 2021). In summary, the number of program phases and the progression criteria are not consistent in the literature. However, a cohort with 131 athletes evidenced that daily assessments of pain, range of motion and muscle strength are useful to inform the progression of rehabilitation (Whiteley et al., 2018), which supports such variables monitoring in the management of HSI. In agreement, all respondents of the current study reported undergoing periodic assessments throughout rehabilitation, being pain (84%) and muscle strength (82%) the most measured variables.

Our findings demonstrate that electrophysical agents are widely used in the rehabilitation of footballers with HSI (98% of respondents), with analgesic currents (93%), photobiomodulation (89%) and cryotherapy (87%) being the most common ones. It is noteworthy that such interventions were not even addressed in the most recent 'Clinical Practice Guideline' (Martin et al., 2022). Electrostimulation (Mendiguchia et al., 2017) and cryotherapy (Malliaropoulos et al., 2004; Sherry & Best, 2004; Silder et al., 2013) were included as adjuvants in a few RCTs. Regarding photobiomodulation, Medeiros et al. (2020) reported no differences between the active and placebo emission of laser therapy on the functional rehabilitation of athletes with HSI. The lack of evidence can also be found for manual therapy and other passive interventions. Massage and myofascial release are manual therapy techniques included in some rehabilitation programs (Jacobsen et al., 2016; Mendiguchia et al., 2017; Tol et al., 2014; Vermeulen et al., 2022; Whiteley et al., 2018) and used by 97% of respondents in the present study. Among the other passive techniques (not related to manual therapy), attention is drawn to the fact that the elastic taping is used by almost half of respondents, even with the ineffectiveness of this intervention evidenced in other musculoskeletal conditions (Ramírez-Vélez et al., 2019).

Resistance exercises to strengthen the hamstrings can be found in almost all studies encompassing HSI rehabilitation (Afonso et al., 2023; Breed et al., 2022) and were mentioned by 98% of respondents of the present study. Exercises using body weight and free weights seem to be the most used, although there was a high percentage reporting other forms of exercise, including isokinetic exercise by more than half of respondents. Since RCTs conducted by Askling et al. (2013, 2014), eccentric strengthening of the hamstrings has been considered a key piece of the HSI rehabilitation programs. More recently, Hickey et al. (2022) evidenced that high-intensity eccentric loading can be safely introduced into early HSI rehabilitation based on exercise-specific progression criteria. Additionally, the Nordic hamstring exercise is an effective strategy to prevent index and recurrent HSIs (Al Attar et al., 2017). Therefore, the adoption of eccentric exercises by 93.5% of the participants is in line with current recommendations (Martin et al., 2022).

A recent scoping review evidenced that exercise interventions in contemporary HSI rehabilitation research are poorly reported (Breed et al., 2022), making it difficult for physical therapists to effectively implement evidence-based protocols in their clinical practice. Unlike more prolonged rehabilitation programs that provide a periodization closer to that recommended for healthy individuals (American College of Sports Medicine, 2009), as in cases of anterior cruciate ligament reconstruction, muscle strengthening following most HSI last just a few weeks. It is also noteworthy that elite football athletes receive daily physiotherapeutic care, often more than once a day. It is a treatment routine different from those implemented by studies involving other profiles of athletes and that perform 2–3 weekly sessions (Askling et al., 2013, 2014; Bayer et al., 2017; Hickey et al., 2020; Medeiros et al., 2020). Even so, the present study reports that most respondents normally use 3–4 sets per exercise, which is consistent with the volume found in several HSI studies (Sherry & Best, 2004; Askling et al., 2013, 2014; Silder et al., 2013; Mendiguchia et al., 2017; Whiteley et al., 2018; Hickey et al., 2020; Medeiros et al., 2020; Vermeulen et al., 2022). Furthermore, a recent 'Clinical Practice Guideline' (Martin et al., 2022) advocated that successful interventions have included 6 to 12 repetitions per set, depending on exercise intensity, with load and range of motion being increased as tolerated. This number of repetitions per set is close to the range of 5–15 repetitions predominantly used by respondents of the current study.

All respondents declared using exercises aimed at increasing range of motion, mainly dynamic stretching (82%). While the flexibility deficit is not evidenced as a risk factor for HSI (Green et al., 2020), the loss of range of motion following a HSI seems to justify the attention to this aspect. Malliaropoulos et al. (2004) demonstrated that increased frequency of static stretching during the rehabilitation of athletes with HSI restores the impaired range of motion faster; thus, stretching is considered an intervention with grade 'B' of recommendation (i.e., moderate evidence) for HSI rehabilitation (Martin et al., 2022). The same grade 'B' of recommendation is assigned to lumbopelvic stabilization exercises (Martin et al., 2022), which are commonly used by 93.5% of respondents. Given the emerging evidence establishing a link between the lumbopelvic kinematics during running and the HSI risk (Franettovich Smith et al., 2017), exercises targeted to 'core' or lumbopelvic stability seem to play a role on HSI prevention (Shield & Bourne, 2018). Since Sherry and Best (2004) showed the benefits of a rehabilitation program including lumbopelvic stabilization exercises this type of intervention has been included in the main HSI studies (Askling et al., 2013, 2014; Hickey et al., 2020; Malliaropoulos et al., 2004; Medeiros et al., 2020; Mendiguchia et al., 2017; Moen et al., 2014; Sherry & Best, 2004; Silder et al., 2013; Vermeulen et al., 2022). Lumbopelvic stabilization exercises have a grade 'B' of recommendation (i.e., moderate evidence) for HSI rehabilitation (Martin et al., 2022), but evidence regarding the best type of exercise and the ideal dosage (or at least the minimal dosage) is still lacking in the literature.

Sherry and Best (2004) also introduced the idea of progressive agility exercises in HSI rehabilitation, which currently achieve a grade 'B' of recommendation (Martin et al., 2022). In the present study, we addressed this type of intervention within the context of exercises that mimic the functional demands of football (running, changing direction, jumping, passing/kicking, etc.) and we verified the high rate of use of this type of stimulus. Interestingly, sprints have been considered the main HSI mechanism in football (Ekstrand et al., 2020) and were not included in the rehabilitation programs of some respondents of our survey. It is possible, however, that other members of the health/coaching staff (e.g., strength and conditioning coaches) are in charge for the transition to the field phase in some clubs. Considering the potential preventive effect of the athletes' exposition to sprints on the risk of HSI (Edouard et al., 2023), the inclusion of this type of stimulus in more advanced phases of rehabilitation programs seems advisable.

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Regarding the assessments carried out for decision-making about RTP, we observed heterogeneity of responses within a range of outcomes. Muscle strength levels (71%) and performance in sprints/high-speed running (58%) are the only outcomes used as criteria for discharge by more than half of respondents. According to a worldwide survey on RTP practices following HSI in premier league football clubs (Dunlop et al., 2020), absence of pain, hamstring strength, training load and functional performance/sport-specific tests were the most frequent outcomes. Our findings and those reported by Dunlop et al. (2020) are partially in line with a Delphi study (van der Horst et al., 2017) in which experts reached a consensus on the following criteria to support the RTP decision following HSI in football athletes: medical staff clearance, absence of pain on palpation, absence of pain during strength and flexibility testing, absence of pain during/after functional testing, similar hamstring flexibility, performance on field testing, and psychological readiness. It is noteworthy that predictive value of such criteria for re-injury rate and its specific cut-offs for professional footballers require further investigation.

Some limitations of this study should be acknowledged. First, considering that only Brazilian physical therapists participated in this research, the results do not necessarily reflect the management provided to high-level men's football athletes with HSI in other countries. Aspects related to the professional training of physical therapists in Brazil may affect the respondents' choices, as well as social, psychological, and cultural aspects of Brazilian football. Therefore, caution is needed when comparing our findings with other football contexts worldwide. Furthermore, considering sex-related issues in rehabilitation of female athletes with HSI (O'Sullivan, Preszler, & Tanaka, 2022), replication of the current study among woman's football clubs is recommendable. Lastly, the data collection instrument (i.e., self-completed questionnaire) is prone to inconsistent responses. We minimized this limitation through direct contact with the participants for clarification after completing the questionnaires, leading to a very low rate of missing data.

5. Conclusion

The present study allowed the sports physical therapy community to become aware of the approaches usually adopted for management of athletes with HSI who play in the highest level of Brazilian men's football. Despite the heterogeneity of choices regarding assessment practices, all respondents use imaging exams, adopt injury classification scales, and evaluate aspects related to pain, range of motion, muscle strength, and functional status of athletes with HSI. Rehabilitation programs are usually divided into 3 to 4 phases. Electrophysical agents, manual therapy, stretching, strengthening exercises (including eccentrics), lumbopelvic stabilization exercises and exercises that mimic the functional demands of football are used by vast majority of respondents. Muscle strength was the most reported RTP criterion.

Ethical statement

The study was approved by the Ethics in Research Committee of the Federal University of Health Sciences of Porto Alegre, Porto Alegre, RS, Brazil (approval number: 4.838.243).

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

Authors declared no conflict of interest.

Acknowledgements

BMB thanks CNPq-Brazil for the research productivity fellowship.

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